Step 3 : Specify the level of significance, α.

Step 4 : Consider the test statistic $Z = \dfrac{p - P}{\sqrt{\dfrac{PQ}{n}}}$ under H_0. Here, $Q = 1 - P$.

The approximate sampling distribution of the test statistic under H_0 is the $N(0,1)$ distribution.

Step 5 : Calculate the value of Z under H_0 for the given data as $z_0 = \dfrac{p - p_0}{\sqrt{\dfrac{p_0 q_0}{n}}}$, $q_0 = 1 - p_0$.

Step 6 : Choose the critical value, z_e, corresponding to α and H_1 from the following table

Alternative Hypothesis (H_1)	$P \neq p_0$	$P > p_0$	$P < p_0$
Critical Value (z_e)	$z_{\alpha/2}$	z_α	$-z_\alpha$

Step 7 : Make decision on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$P \neq p_0$	$P > p_0$	$P < p_0$
Rejection Rule	$\lvert z_0 \rvert \geq z_{\alpha/2}$	$z_0 > z_\alpha$	$z_0 < -z_\alpha$

Example 1.12

A survey was conducted among the citizens of a city to study their preference towards consumption of tea and coffee. Among 1000 randomly selected persons, it is found that 560 are tea-drinkers and the remaining are coffee-drinkers. Can we conclude at 1% level of significance from this information that both tea and coffee are equally preferred among the citizens in the city?

Solution:

Step 1 : Let P denote the proportion of people in the city who preferred to consume tea. Then, the null and the alternative hypotheses are

Null hypothesis: $H_0 : P = 0.5$

i.e., it is significant that both tea and coffee are preferred equally in the city.

Alternative hypothesis: $H_1 : P \neq 0.5$

i.e., preference of tea and coffee are not significantly equal. It is a two-sided alternative hypothesis.

Step 2 : **Data**

The given sample information are

Sample size (n) = 1000. Hence, it is a large sample.

No. of tea-drinkers = 560

Sample proportion (p) = $\dfrac{560}{1000} = 0.56$

Step 3 : **Level of significance**

α = 1%

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